Online Acoustic Camera

User Manual And API

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Revision Information

Version	Date	Editor	Description
V1.0	2022.04.21	Xianfeng Xie	Initial Version
V2.0	2022.08.30	Xianfeng Xie	HTTP, W-S interface newly added
V3.0	2023.01.30	Ziwei Wang	New functions added
V4.0	2023.04.10	Ziwei Wang	New version function updated
V5.0	2023.10.20	Wentao Yang	New version interface updates
V5.1	2024.01.30	Wentao Yang	Update of gas leakage model

1 System Introduction

1.1 Overview

Acoustic camera, also known as acoustic imager, is a device that converts sound fields into visual images. Acoustic imaging is based on microphone array measurement technology, which measures the phase difference of signals from sound waves in a certain space to various microphones. Based on the principle of phased array, the position of the sound source is determined, the amplitude of the sound source is measured, and the spatial distribution of the sound source is displayed in the form of an image, that is, the spatial sound field distribution cloud map - the sound image map is obtained, where the color and brightness of the image represent strength. This device has functions of sound acquisition, image acquisition, and sound localization visualization, achieving visual detection of equipment faults. The device is small in size and lightweight, and can be mounted on various inspection robots, further empowering the robot inspection method.

This system adopts internationally leading ESR environmental sound target recognition technology and sound source localization technology based on beamforming theory. It measures the position of the sound emitted by the measuring equipment and the state of sound radiation, covering the entire range from audible to ultrasonic frequencies. The collected sound is visualized in the form of a color contour map. Effective measurement of sound field distribution, perfect superposition of sound field images and visible light video images, it works quite similar to a thermal imager for detection of object temperature. This device not only has the functions of traditional acoustic camera sound acquisition, image acquisition, and sound localization visualization, but can also detect and locate abnormal positions in real time when the device malfunctions.

This device can be used in various industrial scenarios, such as partial discharge monitoring for high-voltage transmission lines, gas leakage monitoring for high-voltage pipelines, and partial discharge fault monitoring for transformers, switchgear, and other equipment. This system can be used for sound source localization, abnormal sound detection, sound source position tracking and localization, etc. For steady-state or high transient sound sources, both stationary and moving objects can achieve excellent detection results.



Figure 1-1 Partial Discharge Monitoring Scenarios



Figure 1-2 Gas Leakage Monitoring Scenarios

1.2 System Features

Compared with other types of partial discharge monitoring or gas leakage monitoring products, Online Acoustic Camera have the following features:

- Intelligent integrated design of sound and light imaging, with a lightweight appearance
- Intrinsic safety design meets explosion-proof requirements
- Wide detection frequency band, covering both audible and ultrasonic signals
- Realize automatic detection of sound field visualization, intuitive, fast, and accurate positioning
- Supports steady-state continuous detection mode and transient detection mode
- Support detection and imaging of common PD signals

- Support detection and imaging of gas leaks in high-pressure pipelines
- Open interfaces and protocols, supporting secondary development

2 Specifications

2.1 Appearance & Dimensions

Pic.2-1 is the physical appearance image of the products



Figure 2-1 Appearance Diagram

Pic. 2-2 is the Appearance dimension diagram of the Product.

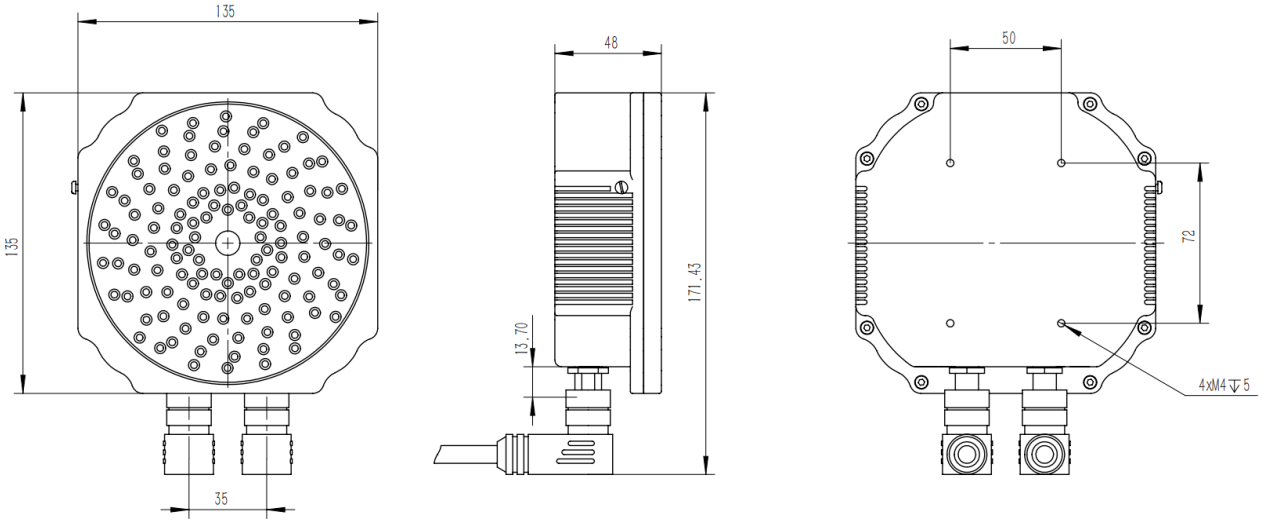


Figure 2-2 Appearance Dimension Diagram

2.2 Performance Index

Parameters	HZ-FA-371

Power supply	DC 12V/3A
Sensors	128
Sensor type	MEMS microphone
Sensor dynamic range	30dB~120dB
Frequency band range	2~65kHz
Measuring distance	0.5m~50m
Application Scenarios	Partial discharge, Gas leaks
Dimension	135mm*135mm*48.5mm
Weight	< 900g
Protection	IP68
Focal length	3.9mm
FOV	60°
Transmission	Ethernet
Resolution	1920*1080
Video Protocol	RTSP

3 System Deployment

3.1 Installation

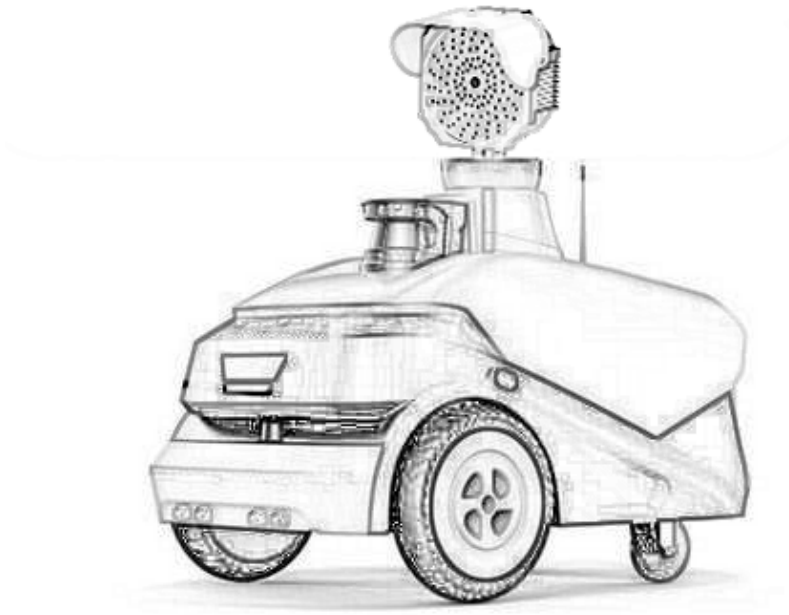


Figure 3-1 A mobile application with acoustic camera

This device is compact and light weight and can be installed on the robot gimbal or other suitable installation positions (as shown in the above figure).

During the installation process, it is important to ensure that the front end of the device is not obstructed by objects, and to minimize the impact of robot vibration on the device to avoid affecting the later detection effect. The installation direction of the device can be determined according to the robot inspection requirements, and the structural components during the installation process need to be prepared by users themselves.

The device can be installed on a fixed place to monitoring the high significant zone to monitoring the Partial discharge and gas leaks.

3.2 Network Topology

The topology diagram of device network connection in a typical scenario is shown in Pic. 3-2. Devices and computers are generally located within the same local area network, and real-time video footage can be transmitted and parameter configured on site through the HTTP protocol.

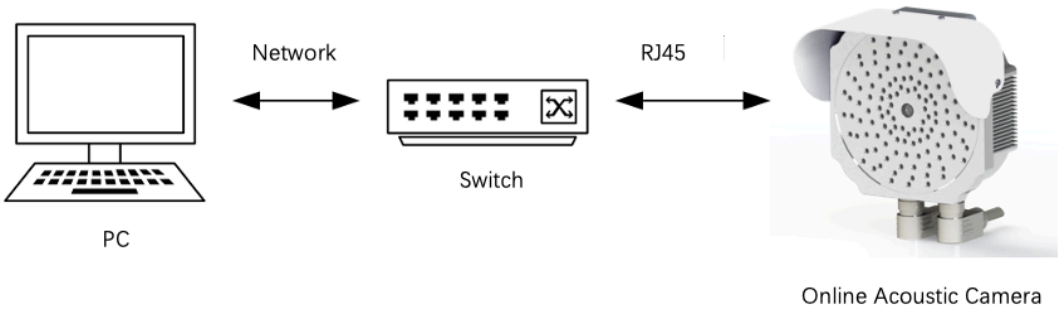


Figure 3-2Network connection topology.

Notes: Default IP for the device is 192.168.2.204, and the computer IP needs to be kept in the same network segment as the device IP.

4 System configuration

4.1 System login

When the network connection is normal, enter the device IP (default: 192.168.2.204， shows on label) in the browser to enter the login page, as shown in Figure 4-1. The default username and password are as follows:

User Name：admin

Password：123456

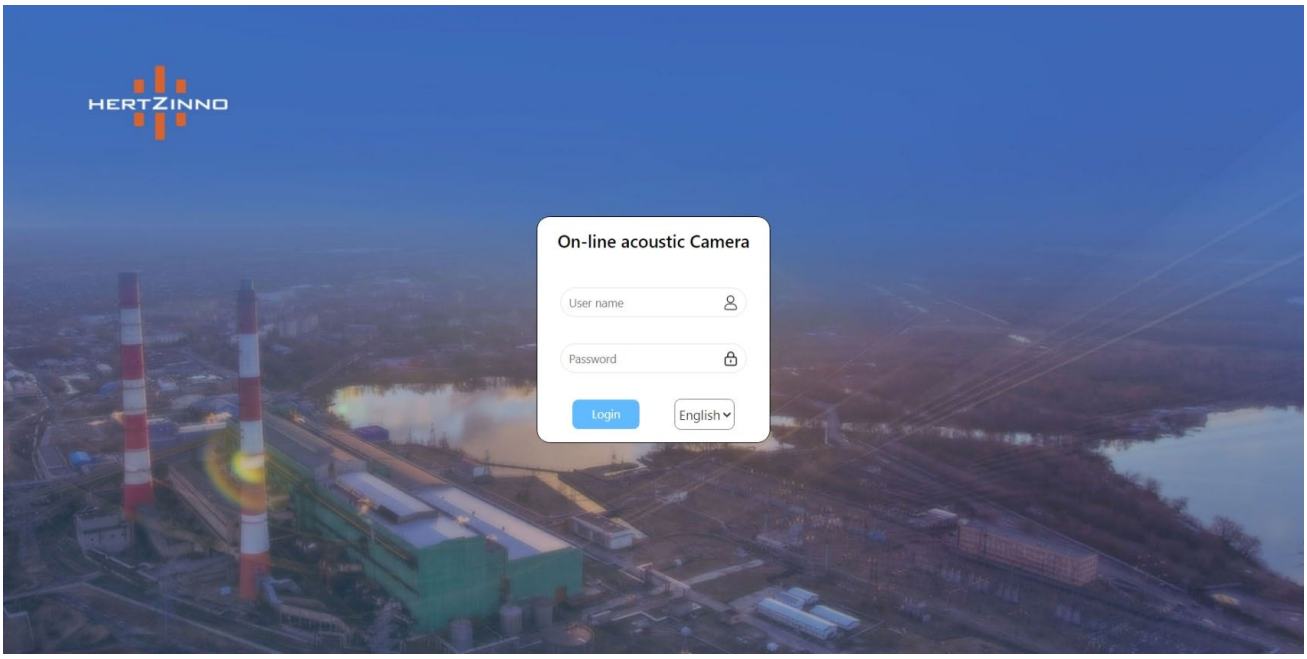


Figure 4-1 Login interface

4.2 Live view Page

After logging in, enter the screen preview interface. The real-time video screen will be displayed on the left side of the page, and the parameter console will be displayed on the right side of the page. The partial discharge detection mode (gas leakage detection mode) in the upper right corner is the current device working mode, and the system supports switching between Chinese and English.

Click the power button ⏻ to log out of the system.

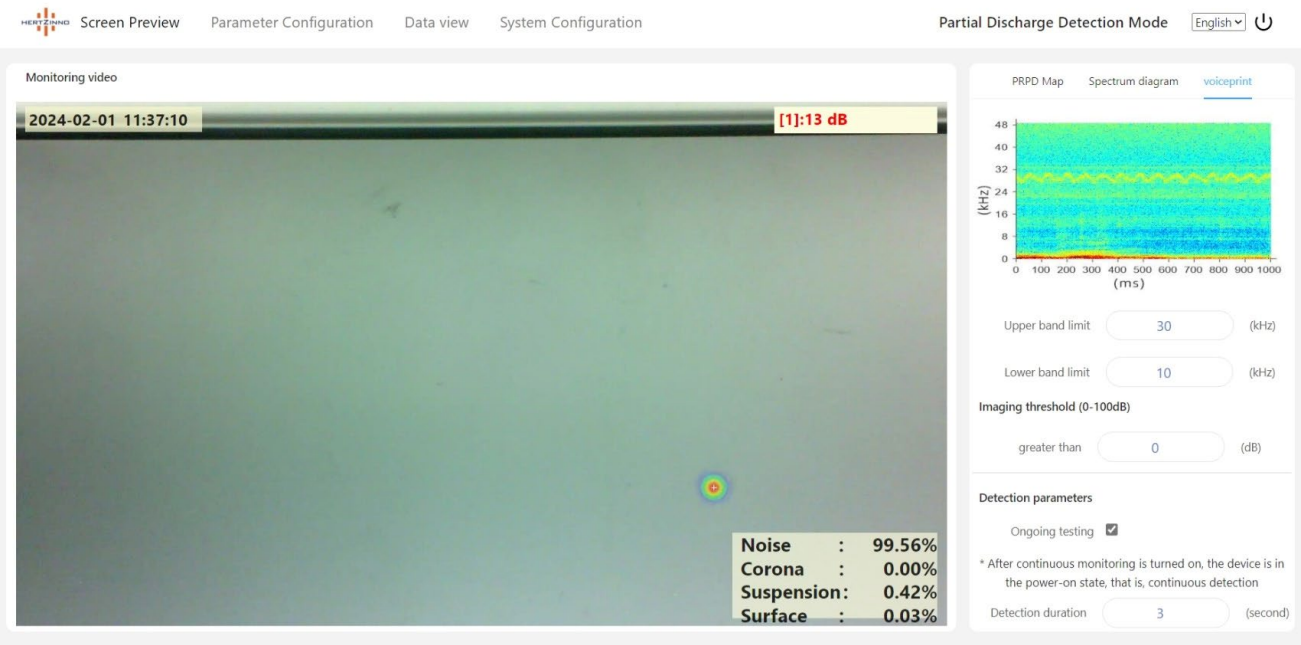


Figure 4-2 Liveview interface

4.2.1 Real-time video

This interface allows real-time viewing of the current video screen and adjustment of parameters on the right console. The upper left corner of the video shows the system time, and the upper right corner displays the real-time environmental sound energy in the current monitoring frequency band. When the environmental energy value in the corresponding frequency band is detected to be higher than the threshold value, an abnormality will be displayed, and a sound field map will appear on the real-time screen, which is the location of partial discharge/air leakage.

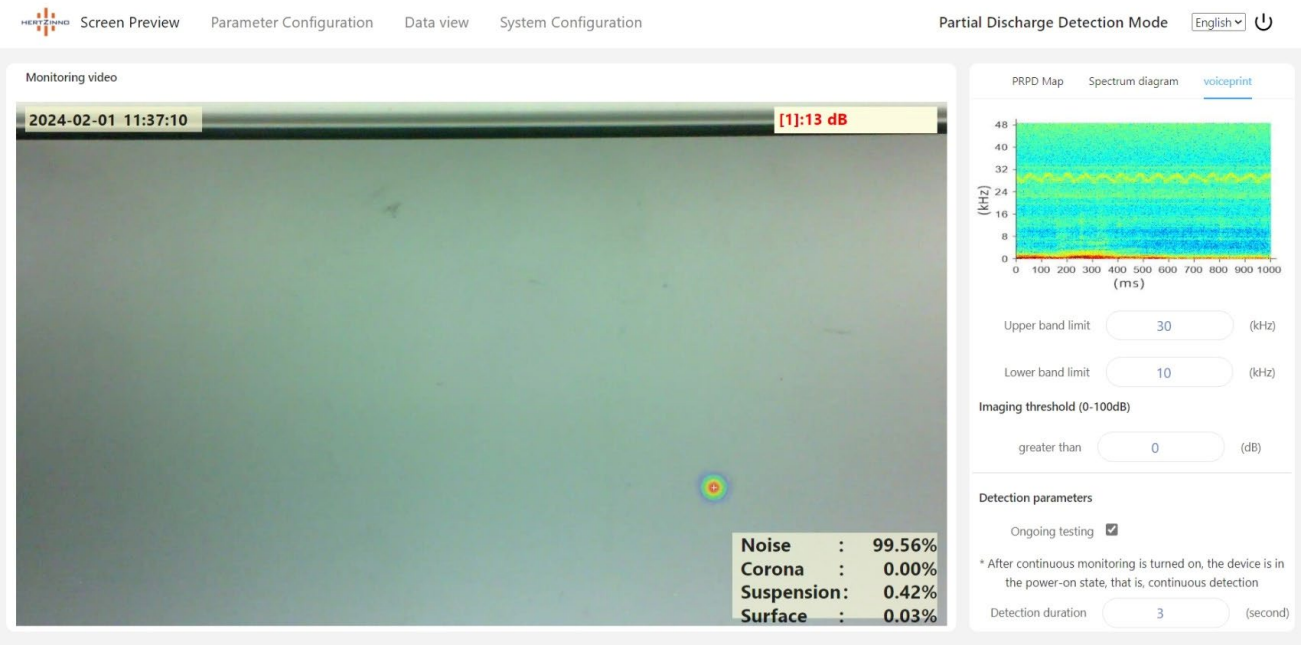


Figure 4-3 liveview image

4.2.2 Console

The functional descriptions of each part of the console are as follows:

- PRPD Diagram: For partial discharge monitoring, real-time analysis and creation of PRPD maps to show partial discharge points.
- Spectrum Diagram: Real time analysis and drawing of the spectrum of the current environmental sound signal. Through the spectrum diagram, the distribution of sound in various frequency bands of the environment can be visually observed, providing reference for imaging frequency band adjustment.
- Voiceprint Diagram: Real time analysis and drawing of the voiceprint of the current environmental sound signal.
- Imaging frequency band: used to adjust the range of the currently monitored sound frequency band. The adjustable range is 0-48kHz(2-65kHz), and the numerical input must be an integer. The difference between the upper and lower limits cannot be greater than 20 or less than 2, and the upper limit must be greater than the lower limit. After adjusting the frequency band, the shaded areas in the spectrogram are the currently selected frequency band range.
- Imaging threshold: used to adjust the detection threshold. When the energy value of the on-site

environment exceeds the threshold value, the state will become abnormal and real-time sound field imaging can be performed. The sound field imaging marking screen can be observed in the video.

4.3 Parameter Configuration

Enter the parameter configuration interface to change parameters such as the number of detection targets and video settings.

4.3.1 Setting the Number of Detection Targets

The system supports single or multiple object detection, available choices are 1/2/3.

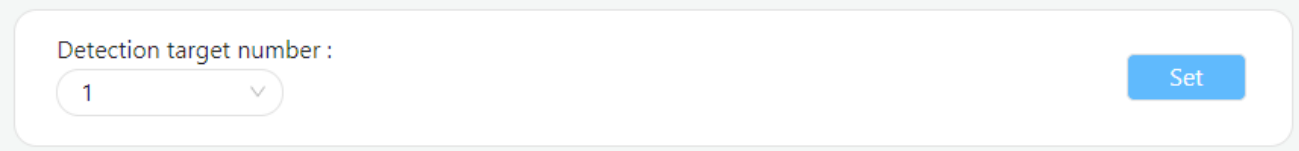


Figure 4-4 Configuration of Detection Target Number

4.3.2 Video Settings

The RTSP service port can be configured, including video resolution, video frame rate, video bit rate, etc. The video encoding defaults to H.264 and cannot be modified. The video resolution is fixed and can be selected as 2592 * 1944, 1920 * 1080, 1280 * 720, 720 * 480, 640 * 480. The input range for video frame rate is [1,30], and the video stream is fixed and can be selected as 4096, 2048, 1024, 512.

If there are any modifications to the above content, you need to click the “Set” button in the upper right corner to take effect after the modification is completed.

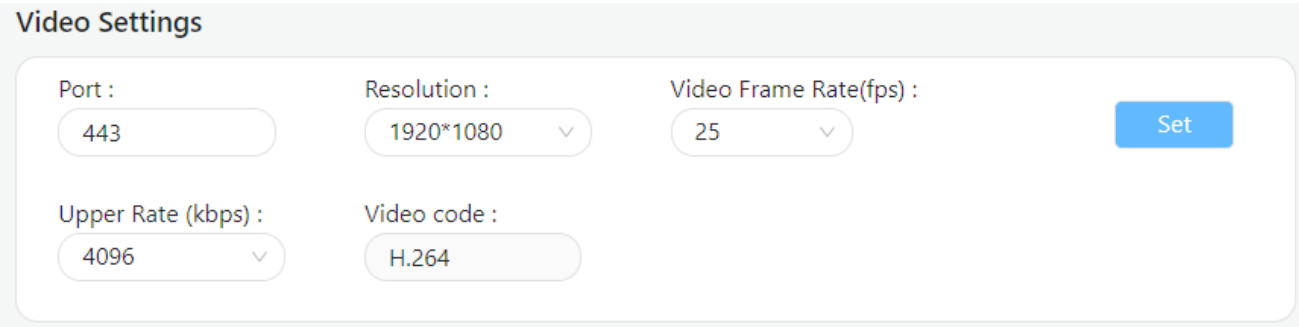


Figure 4-5 Video Parameter Configuration

4.3.3 Other Settings

Other settings are set to the box selection function. First, check the enable custom alarm area below, then click the draw area button above, drag the mouse to select the range to be detected, and click to stop drawing after completion. The area drawn in the figure is the detection area, and the device will only detect the partial discharge situation inside the selected area. If the selected area is not suitable, you can clear the drawn detection box by clicking the clear all button next to it, and then redraw the detection box through the interface.

If there are any modifications to the above content, you need to click the "Set" button in the right volume to take effect after the modification is completed.

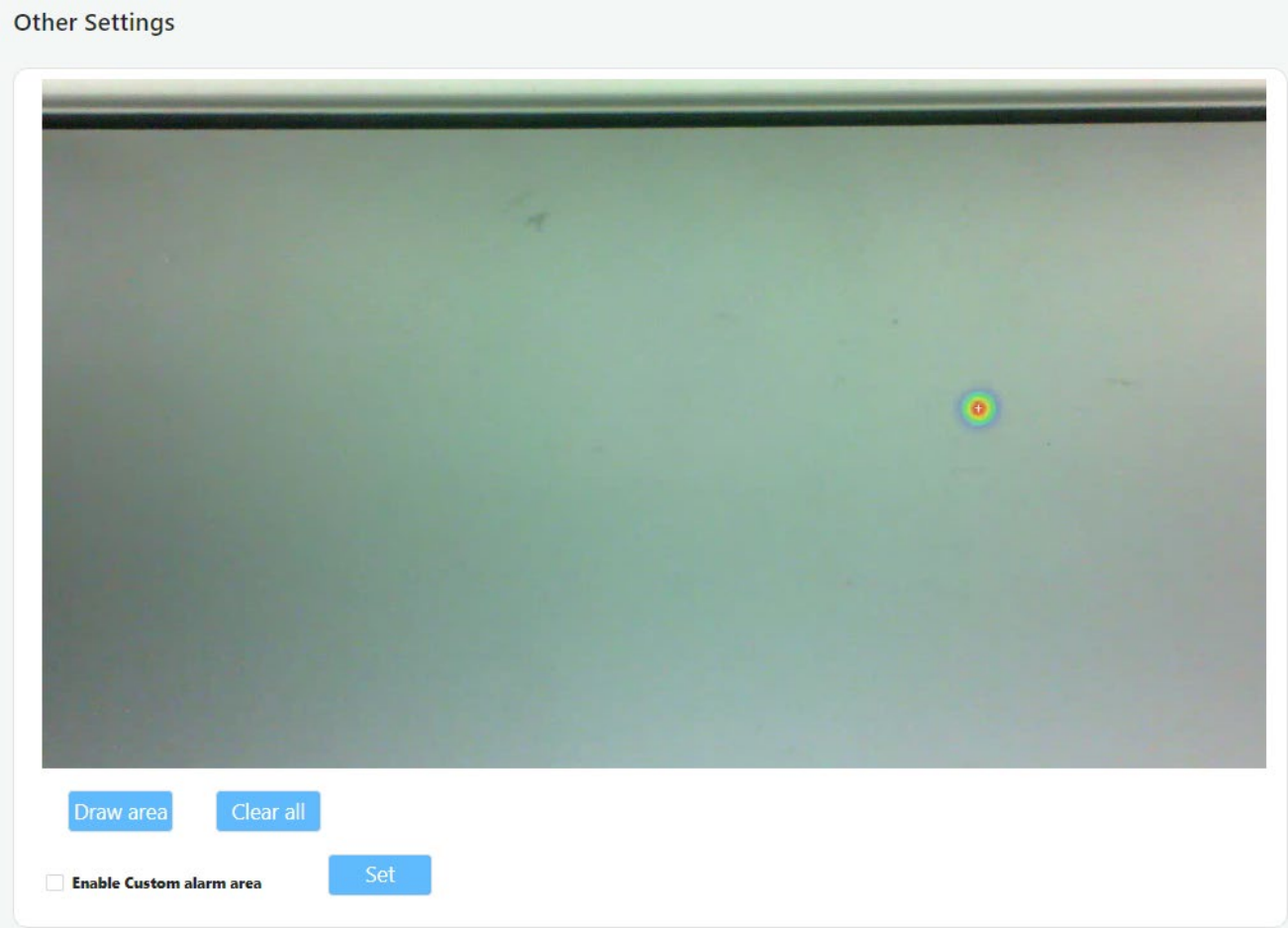


Figure 4-6 Other Parameter Configuration

4.4 Data Viewing

This page mainly records the file information that generates detection records during device operation. The storage space of the device is limited, and only the latest 100 detected file information

will be retained. If it exceeds the limit, the earliest saved record will be deleted, and new data will be recorded to achieve circular storage. On this page, the detection records can be searched and queried based on the detection generation time and discharge type. The characteristic values of the environment during detection can be viewed, and the detection records support viewing and downloading of images, videos, and other content generated during detection.

Signal types are divided into: Normal, Corona Discharge, Surface Discharge, and Floating Discharge.

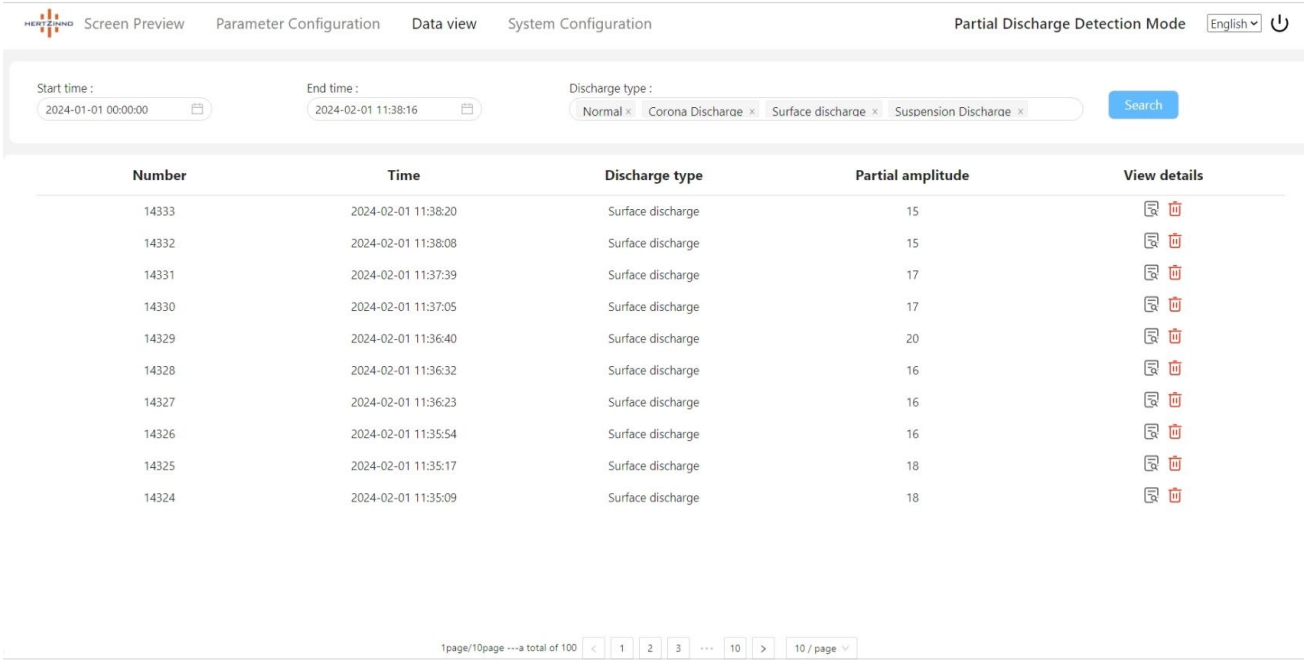


Figure 4-7 Data Viewing Interface

4.5 System Configuration

Enter the system configuration interface to change device time, network, heartbeat, user password, etc.

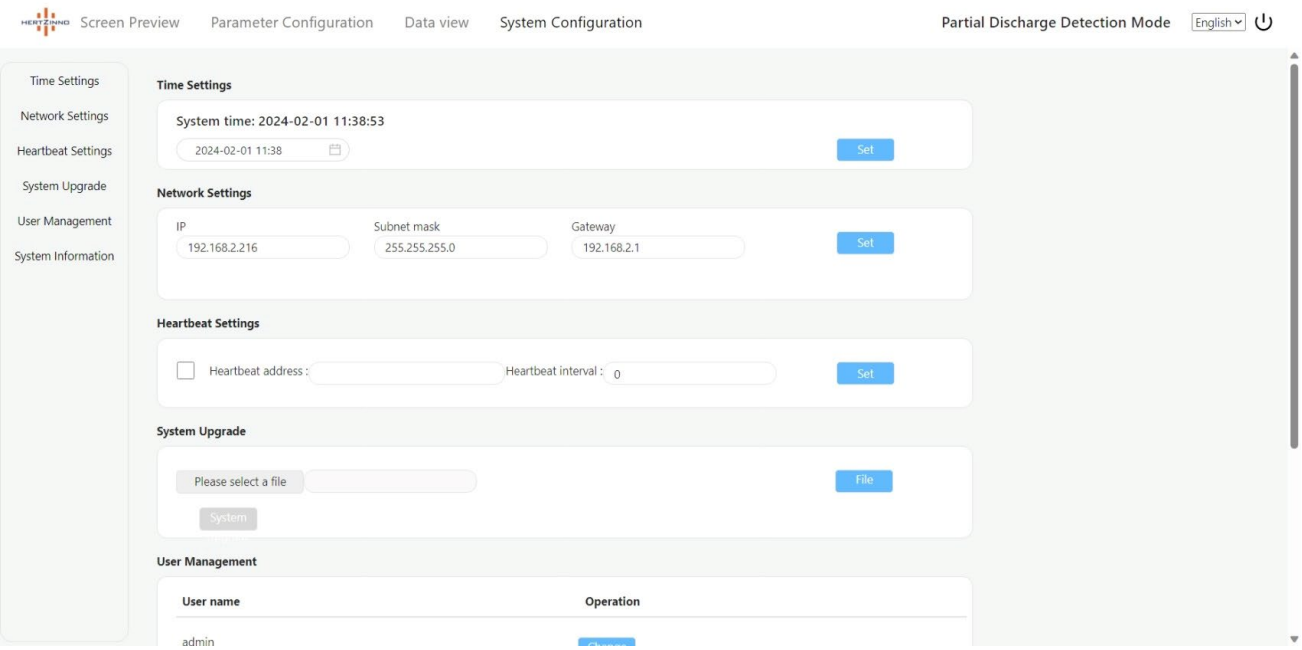


Figure 4-8 System setting interface

4.5.1 Time setting

The system time is the current time of the device. By clicking on the calendar icon , users can set their own device time. Click the refresh icon , and the system will automatically synchronize the computer time to the editing box.

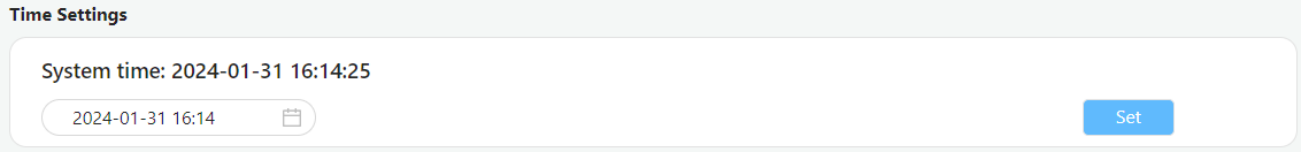


Figure 4-9 Time Setting

4.5.2 Network Configuration

The default IP of the device is 192.168.204. If changes are needed, they can be made in the network configuration. After the modifications are completed, click on Settings. After setting up, the system will automatically log out. You need to re-enter the newly set IP in the browser to enter the system.

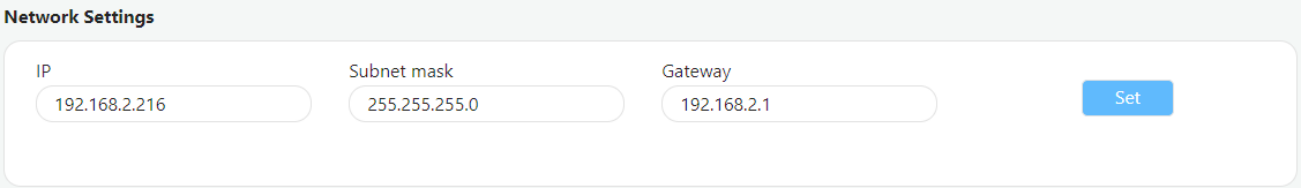


Figure 4-10 Network settings

4.5.3 Heartbeat setting

Return the set heartbeat information to the specified network address at fixed intervals. After checking, the device can return heartbeat information to the specified network address at a set interval time.

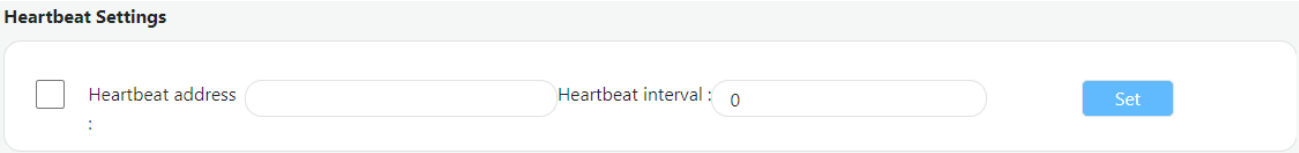


Figure 4-11 Heartbeat setting

4.5.4 System Upgrade

The device supports upgrade. When there is a version update in the software, the obtained update package can be loaded into the device through remote upgrade.

Operation method:

1. Click to select the file and select the update package you want to upgrade;
2. After confirmation, click on file upload again;
3. After successful upload, click on file upgrade to achieve remote software upgrade function.

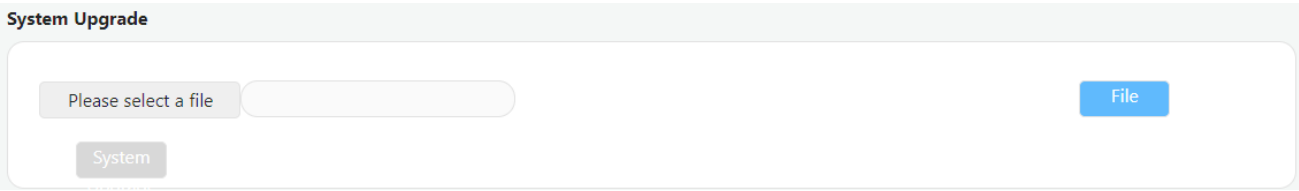


Figure 4-12 System Upgrade

4.5.5 User Management

Click to change password, users can modify their login password by themselves, and confirm after modification.

Default User Name: admin (Not modifiable) Password: 123456

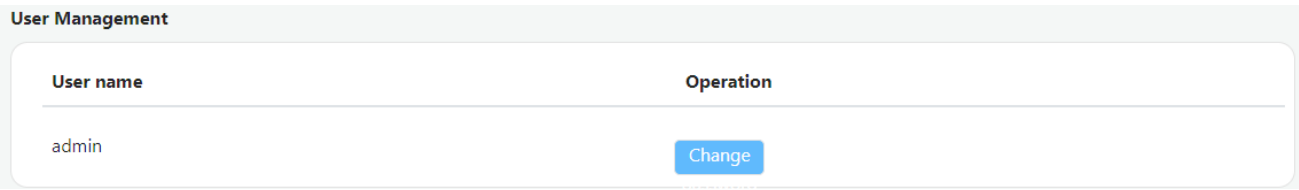


Figure 4-13 User Management

4.5.6 System Information

Display current device information and support custom naming of devices.

Restore: One click to restore default settings, IP, username, password, and other information will be reset to default values and all historical data will be deleted.

Restart device: Click the restart device button to remotely restart the device. After restarting, the system will automatically exit. After the device starts automatically, you need to re-enter the IP address in the browser to enter the system.

Device Logs: Click on Set Logs and the system will automatically download the log file locally, with the file name: system logs. log.



Figure 4-14 System Information

5 Open API Description

5.1 HTTP Interface

5.1.1 Obtain token for authentication.

Function Description: Authentication token distribution interface that returns an encrypted token value by validating the username and password. All subsequent interfaces require the inclusion of a bearer-type token value in the Authorization field of the HTTP Header for authentication. If authentication fails or the token is invalid, a 401 error will be returned.

Request Format: http://localhost/login

Request Method and URL: POST /login

Interface Request Body: Yes

Encrypted Request Example:

{

"username"："admin"，

"password"："**********"

}

Interface Request Parameter Description：

Parameter name	Type	Description
username	string	User name
password	string	“passwordlfxs” MD5 encrypted ciphertext, 32-bit lower case. Eg Password is 123456, the request should be 123456lfxs

The interface returns JSON data as follows：

{

"token"： "**********"

}

5.1.2 Obtaining system configuration

Function Description: Used to obtain the values of parameters such as imaging energy threshold, alarm threshold, upper and lower detection band limits, and device monitoring mode.

Request Format: http://localhost/system

Request method and URL: GET /system

Interface request body: None

The interface returns JSON data as follows：


```
{
    "work_mode": 1,
 "dB_limit": 10,
    "freq_down_limit": 20,
    "freq_up_limit": 24,
    "detect_times": 3,
    "samplerate": 96000
}
```


Interface return parameter description：

Parameter	Type	Required or not	Description
work_mode	integer	YES	**Device monitoring modes** 1: partial discharge detection 2: leakage detection
dB_limit	integer	YES	Imaging energy threshold, range 0-100
freq_up_limit	integer	YES	Analysis band upper limit (unit KHz)
freq_down_limit	integer	YES	Lower analysis band limit (unit KHz)
detect_times	integer	YES	Detection timeout duration (unit second)
samplerate	integer	YES	MIC matrix sampling rate (unit Hz)

5.1.3 Setting the system configuration

Function Setting: used to set the value of parameters such as imaging energy threshold, upper and lower detection band limits, detection timeout duration, etc.

The parameter name is the same as above, when requesting, if you need to configure the imaging energy threshold, then send the 'dB_limit' field separately, and the detection timeout length is configured in the same way. When configuring the analysis frequency band, you need to transmit the upper and lower frequency band limit fields at the same time.

Request format: http://localhost/system

Request method and URL: PUT /system

Interface request body: Yes

Headers:

Parameter	Parameter value	Required or not
Content-Type	application/json	YES

Example: Set the upper and lower limits of the frequency band (the same for other contents of the configuration)

Requirements: 2 ⩽ upper limit - lower limit ⩽ 20

```
{
    "freq_down_limit": 20,
    "freq_up_limit": 30
}
```

Interface Request Parameter Description：

Parameter	Type	Required or not	Description
freq_down_limit	integer	Not	Lower limit of analysed frequency band (unit KHz)
freq_up_limit	integer		Upper limit of analysing frequency band (unit KHz)
detect_times	integer	Not	Detection timeout duration (in seconds)

dB_limit	integer	Not	Imaging energy threshold, range 0-100

5.1.4 Getting the current configuration (video parameters)

Function Description: Used to get the values of parameters such as resolution, video frame rate, video bit rate, RTSP service port and so on.

Request Format: http://localhost/video_config

Request method and URL: GET /video_config

Interface request body: None

Return JSON data as follows:

```
{
    "video_bit_rate": 4096,
    "video_fps": 25,
    "video_height": 1080,
    "video_port": 443,
    "video_width": 1920
}
```

Interface return parameter description：

Parameter	Type	Required or not	Description
video_bit_rate	integer	YES	Video bitrate, take fixed value: 4096, 2048, 1024, 512
video_fps	integer	YES	Video frame rate, range: [1,30]
video_height	integer	YES	Resolution height
video_width	integer	YES	Resolution width
video_port	integer	YES	rtsp service port

Note: The resolution takes a fixed value: 2592*1944、1920*1080、1280*720、720*480、640*480

5.1.5 Setting the current configuration (video parameters)

Function Set the value of resolution, video frame rate, video bit rate, RTSP service port and other

parameters.

Parameter name is the same as above, when requesting, need to configure RTSP service port, then send RTSP service port field separately, video frame rate, video bit rate mode is the same. When configuring the resolution, you need to transmit the resolution high and resolution wide fields at the same time.

Request format: http://localhost/video_config

Request method and URL: PUT/video_config

Interface request body: YesHeaders：

Parameter	Parameter value	Required or not
Content-Type	application/json	YES

Example: When setting the resolution, it is necessary to transmit the resolution height and resolution width fields at the same time according to a fixed value.

```
{
    "video_height": 1080,
    "video_width": 1930
}
```

Description of interface request parameters

Parameter	Type	Required or not	Description
video_height	integer	Not	The height resolution
video_width	integer	Not	The width resolution

The resolution takes a fixed value:2592*1944、1920*1080、1280*720、720*480、640*480

5.1.6 Configuration related to obtaining leakage estimates (in leakage mode only)

Function Description: Used to get the values of the parameters used in leakage estimation.

Request Format: http://localhost/leakage_estimation

Request method and URL: GET /leakage_estimation

Interface request body: None

Return JSON data as follows:

```
{
    "cost_display": 0,
    "leakage_distance": 0,
    "leakage_unit": 0,
    "leakage_unit_str": "l/min",
    "monetary_unit": 0,
    "monetary_unit_str": "CNY",
"energy_cost": 0,
    "specific_power": 0,
"value_of_gas": 0,
    "working_hours": 0
}
```


Interface Request Parameter Description：

Parameter	Type	Required or not	Description
cost_display	integer	YES	Whether to display the results of the leakage cost estimation on the web site, 0 - not displayed, 1 - displayed
leakage_distance	integer	YES	Leakage estimation distance, 0--0.5m, 1--1m, 2--2m, 3--5m
leakage_unit	integer	YES	Leakage estimation and display unit, 0--l/min, 1--ml/s
leakage_unit_str	string	YES	Leakage estimation and display units, l/min or ml/s
monetary_unit	integer	YES	Leakage and loss estimation using currency units, 0 -- ¥ (CNY) [RMB], 1 -- £ (GBP) [Pound Sterling], 2 -- $ (USD) [US Dollar], 3 -- € (EUR) [Euro], 4 -- ₽ (RUB) [Ruble].
monetary_unit_str	string	YES	Leakage and damage estimation using strings of monetary units, including the five units

			mentioned above - CNY, GBP, USD, EUR, RUB
value_of_gas	float	YES	Gas value in monetary units per m3.
energy_cost	float	YES	Energy cost in currency units/kWh
specific_power	float	YES	Specific power in kW/(m3/min)
working_hours	integer	YES	Average annual working hours in h/year

5.1.7 Setting up leakage estimation-related configuration (in leakage mode)

Function Set the value of a parameter used in leakage estimation.

Parameter name is the same as above, when requesting, need to configure a certain parameter, then send the corresponding field separately.

Request Format: http://localhost/leakage_estimation

Request method and URL: PUT /leakage_estimation

Interface request body: Yes

Headers：

Parameter	Parameter value	Required or not
Content-Type	application/json	YES

eg：


```
{
    "cost_display": 0,
    "leakage_distance": 0,
    "leakage_unit": 0,
    "monetary_unit": 0,
"energy_cost": 0,
    "specific_power": 0,
"value_of_gas": 0,
    "working_hours": 0
```

}

Interface request parameter description

Parameter	Type	Required or not	Description
cost_display	integer	Not	Whether to display the leakage cost estimation result on the web end, 0 - not display, 1 - display
leakage_distance	integer	Not	Leakage estimation distance, 0 - 0.5 metres, 1 - 1 metre, 2 - 2 metres, 3 - 5 metres
leakage_unit	integer	Not	Leakage estimation and display unit, 0 - l/min, 1 - ml/s
monetary_unit	integer	Not	Leakage and loss estimation using currency units, 0 -- ¥ (CNY) [RMB], 1 -- £ (GBP) [Pound Sterling], 2 -- $ (USD) [US Dollar], 3 -- € (EUR) [Euro], 4 -- ₽ (RUB) [Ruble]
value_of_gas	float	Not	Gas value, in: currency units/m3
energy_cost	float	Not	Energy cost in currency units/kWh
specific_power	float	Not	Specific power in kW/(m3/min)
working_hours	integer	Not	Average annual working hours in h/year

5.1.8 Get detection area

Function Description: Used to get the value of the detection area parameter.

Request format: http://localhost/signal_filtering_area

Request method and URL: GET /signal_filtering_area

Interface request body: None

The interface returns JSON data as follows:

{

"area_height": -1,

"area_width": -1,

"detect_area_status": 0,

"left_top_x": -1,

"left_top_y": -1

}

Interface return parameter description：

Parameter	Type	Required or not	Description
area_height	integer	YES	Height of the detection area box
area_width	integer	YES	Detect the width of the area box
detect_area_status	integer	YES	Detect area drawing state, 0 hide 1 show
left_top_x	integer	YES	Horizontal coordinate of the upper left corner
left_top_y	integer	YES	Vertical coordinate of the upper-left corner

5.1.9 Setting the detection area

Used to set the value of the detection area parameter, the name of the parameter is the same as above, all fields need to be sent when requesting.

Request format: http://localhost/signal_filtering_area

Request method and URL: PUT /signal_filtering_area

Interface request body: Yes

Headers：

Parameter	Parameter value	Required or not
Content-Type	application/json	YES

Sample interface request parameters:

{

"area_height": -1,

"area_width": -1,

"detect_area_status": 0,

"left_top_x": -1,

"left_top_y": -1

}

5.1.10 Request to enable detection

Functional description: for starting the detection task

Request Format: http://localhost/turn_on_detect

Request method and URL: POST /turn_on_detect

Interface request body: none

5.1.11 Request to end the detection

Function: used to end the detection task (currently all the detection results are transmitted asynchronously via websocket, so the end of the detection request does not return content, see details in 5.2.4)

Request Format: http://localhost/turn_off_detect

Request method and URL: POST /turn_off_detect

Interface request body: None

5.1.12 Getting the current detection status (whether or not detection is in progress)

Function Description: Used to get the detection status

Request Format: http://localhost/detect_status

Request method and URL: GET / detect_status

Return JSON data as follows:

{

"detect_status": 0

}

Interface return parameter description：

Parameter	Type	Required or not	Description
detect_status	boolean	YES	Whether detection is in progress 0: No detection is in progress 1: Detection is in progress

5.1.13 Obtaining detection logs (Partial discharge detection mode)

Function description: used to get the results of the inspection task

Request format: http://localhost/detect logs

Request method and URL: GET /detect_logs

Interface request body: request parameters need to be put into the Query Param of the HTTP request, the relevant parameters are as follows.

Interface Request Parameter Description:

Parameter	Requested or not	Default	Description
start_col	NOT	1	Starting rows
col_count	NOT	100	Number of fetches
start_time	NOT	0	Start timestamp
end_time	NOT	2000000000	End timestamp
discharge_type	NOT	15	Collection of discharge types to be searched
detect_mode	YES	1	Detection Mode: Default is 1, which means searching for logs in local discharge detection mode.

Notes:

1, the system has a total of 60 data, such as the need to start from 50 to get a total of 5, then start_col is 50, col_count is 5.

2、discharge_type is a gathering of status fields.

The meaning of the value of the status field --- 1 on behalf of the noise, 2 on behalf of the corona discharge, 4 on behalf of the surface discharge, 8 on behalf of the floating discharge.

If you want to search results for corona discharge or surface discharge, then the discharge_type assignment 2 + 4 = 6;

if you want to search for corona discharge, surface discharge or floating discharge type of discharge records, then the discharge_type assignment 2 + 4 + 8 = 14.

The interface returns JSON data as follows:


```
{
    "logs": [
        {
            "id": 1,
"detect_mode": 1,
            "start_time": "20230407_143129",
            "end_time": "20230407_143154",
"status": 1,
"soundmap_intensity": 46,
"noise_prob": 1,
            "corona_prob": 0,
"floating_prob": 0,
"surface_prob": 0,
            "files_status": 1,
            "saving_path": "2023_04_07_14_31_29_133/",
            "power_spect_photo_name": "power_spectrum_2023_04_07_14_31_29_133.jpg",
            "power_spect_video_name": "power_spectrum_2023_04_07_14_31_29_133.mp4"
              "prpd_photo_name": "prpd_2023_04_07_14_31_29_133.jpg",
            "prpd_video_name": "prpd_2023_04_07_14_31_29_133.mp4"
```

"soundmap_photo_name": "sound_2023_04_07_14_31_29_133.jpg",

"soundmap_video_name": "sound_2023_04_07_14_31_29_133.mp4"

"voiceprint_photo_name": "voiceprint_2023_04_07_14_31_29_133.jpg",

"voiceprint_video_name": "voiceprint_2023_04_07_14_31_29_133.mp4"

}

],

"logs_count": 1

}

The interface returns a description of the parameters:：

Parameters	type	Description
logs_count	int	Total number of logs
logs	array	Return log group, the specific content structure as above example
id	int	Log number
detect_mode	int	Detection mode: 1-Partial discharge detection mode
start_time	string	Detection start time
end_time	string	Detection end time
status	int	Discharge judgement result flag bit: 1 - noise (no discharge), 2 - corona discharge, 4 – surface discharge, 8 - floating discharge
soundmap_intensity	int	Amplitude of local discharge
noise_prob	float	Probability that the test judgement is noise
corona_prob	float	Probability of corona discharge in this test.
floating_prob	float	Probability that the test determines that it is a floating discharge.

surface_prob	float	Probability that the test determines that it is a surface discharge.
file_status	int	File saving status: 0 - deleted 1 - saving
saving_path	string	Start time named folder record file save path
power_spect_video_name	string	Identifies the name of the spectrogram video
power_spect_photo_name	string	Identifies the name of the spectrogram image
prpd_video_name	string	Identifies the prpd plot video name
prpd_photo_name	string	Identify prpd image name
soundmap_video_name	string	Identify the sound field video name
soundmap_photo_name	string	Identify the sound field imaging picture name
voiceprint_video_name	string	Identify the sound pattern video name
voiceprint_photo_name	string	Identify the sound pattern picture name

5.1.14 Obtaining Detection Logs (Leak Detection Mode)

Function description: used to get the results of the inspection task

Request format: http://localhost/detect logs

Request method and URL: GET /detect_logs

Interface request body: request parameters need to be put into the Query Param of the HTTP request, the relevant parameters are as follows.

Interface Request Parameter Description:

Parameters	Type	Description	Parameters
start_col	NOT	1	Starting rows
col_count	NOT	100	Number of fetches
start_time	NOT	0	Start timestamp
end_time	NOT	2000000000	End Timestamp

discharge_type	NOT	3	Collection of leaks to be searched
detect_mode	YES	2	Detection Mode: Default is 2, which represents the search for logs in leak detection mode.

Note:

1, for example, the system has a total of 60 data, you need to start from 50 to get, a total of 5, then start_col is 50, col_count is 5.

2、discharge_type is the gathering status of the logs.

The meaning of the value of the status field --- 1 on behalf of no leakage, 2 on behalf of leakage. If you want to search results for the results of leakage, then discharge_type assigned 2; if you want to search all records, then discharge_type assigned 1 + 2 = 3.

interface returns JSON data as follows：


```
 {
     "logs": [
         {
             "id": 1,
"detect_mode": 2,
             "start_time": "20230407_143129",
             "end_time": "20230407_143154",
"status": 1,
"soundmap_intensity": 46,
"leakage_value": 0,
"leakage_unit": "l/min",
"leakage_cost": 0,
"cost_unit": "CNY",
             "files_status": 1,
```

```
            "saving_path": "2023_04_07_14_31_29_133/",
            "power_spect_photo_name": "power_spectrum_2023_04_07_14_31_29_133.jpg",
            "power_spect_video_name": "power_spectrum_2023_04_07_14_31_29_133.mp4"
"soundmap_photo_name": "sound_2023_04_07_14_31_29_133.jpg",
            "soundmap_video_name": "sound_2023_04_07_14_31_29_133.mp4"
            "voiceprint_photo_name": "voiceprint_2023_04_07_14_31_29_133.jpg",
            "voiceprint_video_name": "voiceprint_2023_04_07_14_31_29_133.mp4"
        }
    ],
    "logs_count": 1
}
```


Description of interface return parameters：

Parameters	Type	Description
logs_count	int	Log Total
logs	array	Return a log group, and the specific content structure is as shown in the example above.
id	int	Log Number
detect_mode	int	Detection Mode: 2 - Leak Detection Mode
start_time	string	Detection Start Time
end_time	string	Detection End Time
status	int	Leak Determination Result Flag: 1 - No Leak, 2 - Leak Detected
soundmap_intensity	int	Leak Magnitude
leakage_value	float	Estimated Leak Volume

leakage_unit	string	Estimated Leak Volume Unit, l/min or ml/s
leakage_cost	float	Estimated Loss Result, in currency units/year
cost_unit	string	String indicating the currency unit for leak and loss estimation, including the five units mentioned above - CNY, GBP, USD, EUR, RUB
file_status	int	File Save Status: 0 - Deleted, 1 - Saving
saving_path	string	Start Time Naming Folder Records File Save Path
power_spect_video_name	string	Identifier for Spectrogram Video Name
power_spect_photo_name	string	Identifier for Spectrogram Image Name
soundmap_video_name	string	Identifier for Acoustic Imaging Video Name
soundmap_photo_name	string	Identifier for Acoustic Imaging Image Name
voiceprint_video_name	string	Identifier for Acoustic Pattern Video Name
voiceprint_photo_name	string	Identifier for Acoustic Pattern Image Name

5.1.15 Get Detection Result File

Function Description: Used to obtain the file for detection results.

Request Format: http://localhost/detect_file/{log.saving_path}/{logs.file_name}

Request Method and URL: GET /detect_file/{log.saving_path}/{logs.file_name}

Interface Request Body: None

Interface Request Parameter Example (using the log result returned in example 5.1.12):

http://192.168.2.204/detect_file/2023_04_07_14_31_29_133/sound_2023_04_07_14_31_29_133.jpg

Interface Response: Returns the corresponding video or image file.

5.1.16 Delete Detection Logs

Function Description: Used to delete the results of a detection task.

Request Format: http://localhost/detect_logs

Request Method and URL: DELETE /detect_logs

Interface Request Body: Yes

Headers：

Parameter	Value	Required or not
Content-Type	application/json	YES

Sample interface request parameters:

```
{

    "id": [" 2 , 3 "]

}
```

Interface Request Parameter Example:

Parameter Name	Type	Description	Required
id	string	Array of IDs for the logs to be deleted	Not

5.2 WebSocket interface

To ensure real-time data transmission, the Websocket protocol is used to obtain device parameters on port 80.

After the connection is established, device parameters are retrieved based on a custom communication protocol.

Request Format: ws://localhost:port

To ensure a correct and secure WebSocket connection, a heartbeat mode is set. After connecting to the device's WebSocket server, all normally used clients must send any information to the server every minute (content is not limited) to ensure that the connection is not interrupted.

Otherwise, the server will determine that the client is not in use and forcibly terminate the connection.

5.2.1 Get fft Data

Request Address: ws://localhost:port/fftData

The device will return FFT data five times per second (the first half of the FFT, length 512). The returned data is in the form of a 16-bit integer binary stream, which can be parsed by byte order.

5.2.2 Get PRPD Image

Request Address: ws://localhost:port/prpdData

5.2.3 The device will return one frame of PRPD image data per second. The returned data is in the form of an integer binary stream, which can be parsed by byte order.

5.2.4 Get Voiceprint Image

Request Address: ws://localhost:port/voiceprintData

The device will return one frame of voiceprint image data per second. The returned data is in the form of an integer binary stream, which can be parsed by byte order.

5.2.5 Get Detection Result

Request Address: ws://localhost:port/detectFeedback

After each detection, the device will return a packet of detection result data. The returned data is in the form of a JSON string, which can be parsed by byte order.

1. For the JSON data returned in the partial discharge detection mode interface:


```
{

"detect_mode": 1,

                "start_time": "20230407_143129",

                "end_time": "20230407_143154",

"status": 1,

"soundmap_intensity": 46,

"noise_prob": 1,

                "corona_prob": 0,

"floating_prob": 0,

"surface_prob": 0,

                "saving_path": "2023_04_07_14_31_29_133/",

                "power_spect_photo_name": "power_spectrum_2023_04_07_14_31_29_133.jpg",

                "power_spect_video_name": "power_spectrum_2023_04_07_14_31_29_133.mp4"

                   "prpd_photo_name": "prpd_2023_04_07_14_31_29_133.jpg",

                "prpd_video_name": "prpd_2023_04_07_14_31_29_133.mp4"

"soundmap_photo_name": "sound_2023_04_07_14_31_29_133.jpg",

                "soundmap_video_name": "sound_2023_04_07_14_31_29_133.mp4"

                "voiceprint_photo_name": "voiceprint_2023_04_07_14_31_29_133.jpg",

                "voiceprint_video_name": "voiceprint_2023_04_07_14_31_29_133.mp4"

}
```


Description of interface return parameters:

Parameter	Type	Description

detect_mode	int	Detection mode：1-Partial Discharge Detection
start_time	string	The start time of detection
end_time	string	The end time of detection
status	int	Discharge determination result flag (1 for noise, 2 for corona discharge, 4 for surface discharge, 8 for floating discharge).
soundmap_intensity	int	Energy of Partail discharge
noise_prob	float	Probability of determining the detection as noise.
corona_prob	float	Probability of determining the detection as corona discharge.
floating_prob	float	Probability of determining the detection as suspended discharge.
surface_prob	float	Probability of determining the detection as surface discharge.
saving_path	string	Folder name recorded with the start time for file saving path.
power_spect_video_name	string	Identifier for the spectrum video name.
power_spect_photo_name	string	Identifier for the spectrum image name.
prpd_video_name	string	Identifier for the PRPD video name.
prpd_photo_name	string	Identifier for the PRPD image name.
soundmap_video_name	string	Identifier for the acoustic imaging video name.
soundmap_photo_name	string	Identifier for the acoustic imaging image name.
voiceprint_video_name	string	Identifier for the voiceprint video name.
voiceprint_photo_name	string	Identifier for the voiceprint image name.

2.The leak detection mode interface returns JSON data as follows:：

{

"detect_mode": 2,

```
                "start_time": "20230407_143129",

                "end_time": "20230407_143154",

"status": 1,

"soundmap_intensity": 46,

"leakage_value": 0,

"leakage_unit": "l/min",

"leakage_cost": 0,

"cost_unit": "CNY",

                "saving_path": "2023_04_07_14_31_29_133/",

                "power_spect_photo_name": "power_spectrum_2023_04_07_14_31_29_133.jpg",

                "power_spect_video_name": "power_spectrum_2023_04_07_14_31_29_133.mp4"

"soundmap_photo_name": "sound_2023_04_07_14_31_29_133.jpg",

                "soundmap_video_name": "sound_2023_04_07_14_31_29_133.mp4"

                "voiceprint_photo_name": "voiceprint_2023_04_07_14_31_29_133.jpg",

                "voiceprint_video_name": "voiceprint_2023_04_07_14_31_29_133.mp4"

}
```


The interface returns parameter descriptions:

Parameter	Type	Description
detect_mode	int	Detection Mode: 2-Leak Detection Mode
start_time	string	Detection start time
end_time	string	Detection end time
status	int	Leakage judgement result flag bit: 1 - no leakage, 2 - leakage
soundmap_intensity	int	Leakage amplitude

leakage_value	float	Leakage estimation
leakage_unit	string	Leakage estimation unit, l/min or ml/s
leakage_cost	float	Leakage loss estimation results in monetary units/year
cost_unit	string	Leakage and loss estimates in strings of currency units, including the above five units - CNY, GBP, USD, EUR, RUB
saving_path	string	Start time named folder records file save path
power_spect_video_name	string	Identifies the name of the spectrogram video
power_spect_photo_name	string	Identifies the spectrogram image name
soundmap_video_name	string	Identifies the sound field imaging video name
soundmap_photo_name	string	Identifies the sound field imaging picture name
voiceprint_video_name	string	Video name for sound pattern
voiceprint_photo_name	string	Name of the image that identifies the sound pattern